A Covariant Screened Theory for a Dark-Energy-Tied Galaxy Law: a Khronon Parent with One Stiffness Term, Healthy in Its Literal Form, Clean in the Solar System, with Fitted Constants and Pending Cosmology

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Abstract

The owner's programme studies a gravity framework whose galactic law is $g_N = g^2/(g+a_0)$, with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ fitted; its extra gravitating mass is a collisionless "cold energy" medium, not a particle. The companion note PAPER58 found that only a screened version, with a stiffness length $L_\star$, passes both Cassini and the galactic law. This note writes that screened theory covariantly and tests it. Every lane froze its criteria before computing and ran a mutation control; both a_0 footings (9.36×10^{-11} and $1.13\times10^{-10}\,\mathrm{m\,s^{-2}}$) are reported and never pooled.

1. **The action.** The Blanchet–Skordis Khronon theory in its literal form (no extra curvature term, $\beta = 0$; no kinetic term for the spatial covector, $\eta = 0$; Blanchet–Skordis $K(Q)$ dust), with an independent spatial covector $\mathbf{P}$ and one stiffness term $-(\gamma_L/2)(D\cdot P)^2$. Its static limit is AQUAL with $\mu = x/(1+x)$ exactly. The free function that gives this lies outside Blanchet and Skordis's admissible class; that choice, and the stiffness term, are postulated. AeST reproduces everything except the stiffness term, which is identically zero there. The record's earlier action (DSF-21 A33) is this theory with $\eta > 0$, which is ill-posed.
2. **Health and well-posedness.** The literal form has no ghost and no gradient instability on the backgrounds tested; only the two tensor modes propagate, at c. Its leaf equations are those of one strictly convex functional, so the instantaneous sector is well posed (weak field). It violates the criterion "no influence faster than light" by a real signal of about $1.2\times10^{-46}\,\mathrm{m\,s^{-2}}$, and meets the criterion of a single global time.
3. **Tests.** Galaxies: the law itself. Cassini: all 24 cells pass, by 4.8×10^{7} to 5.1×10^{9} (the frozen rule reads UNDECIDED on one float-cancellation check). Planets: at most 0.17 of a bound. Outer globulars: the unscreened law is excluded in all 18 cells, which sets a lower bound on $L_\star$ (21.68 / 22.85 pc with the stellar-population M/L) but no finite $L_\star$ beats Newton. Binary pulsars and GW170817: no constraint. Dwarf satellites: a +0.89 dex gap whose supply is only conditionally met. ALMA: expected unscorable.
4. **Fitted, not derived:** κ, $L_\star$ and the dust abundance.
5. **Pending, stated plainly:** the native CMB (a patched Boltzmann code is validated but no native spectrum has been run; it can test only the dust sector and the cosmological G ratio, because a_0 is absent at linear order) and the native cosmic-shear run (running, no verdict). Gaia DR4 is the sealed decisive test: screened $\gamma_v - 1 \approx 8.1\times10^{-6}$ against $\gamma_v \approx 1.17$ for the unscreened reading.

1 Setting and rules

- $\kappa = \frac{1}{2}$ is fitted. The thermal value $\kappa = Z/4\pi$ is Milgrom's relation $2\pi a_0 \approx c^2(\Lambda/3)^{1/2}$ (arXiv:1605.07458, eq. 5), a postulate here (PAPER58). Footings: canonical $a_0 = 9.36\times10^{-11}$, alternative $1.13\times10^{-10}\,\mathrm{m\,s^{-2}}$, never pooled.
- Status words follow the record: DERIVED (from listed premises), COMPUTED, POSTULATED, FITTED, OPEN, PENDING. The source document is `covariant_push_2026_10_22/THEORY_V3_DRAFT.md` (commit 5d7308f3a2); every claim below cites its lane and commit. Lanes C1–C8 are in `covariant_push_2026_10_22/`; D10 and E6 in `doors_push_2026_10_22/`.
- Table 1 lists the lanes used here that are not in PAPER58.

2 The action (C1, C3, C4, C5)

The Blanchet–Skordis Khronon action is $S = (c^3/16\pi G)\int\sqrt{-g}\,[R - 2J(Y) + 2K(Q)]$, with a preferred foliation τ, khronon acceleration A_μ and $Y = A\cdot A/c^4$ (Blanchet & Skordis 2024, 2025; the equations were read via the arXiv HTML and re-derived symbolically in C4, not verified on disk). The screened theory is:

- **Literal form.** $\beta = 0$ (no extra curvature term), $\eta = 0$ (no kinetic term for $\mathbf{P}$), Blanchet–Skordis $K(Q)$ dust.

Table 1: Lanes of the covariant push and two late doors lanes. Commits: criteria / result.

lane	question	verdict	checks	commits
C1	AeST embedding	EMBEDS WITH EXTRA TERM	8/8	d9a9a17f4e / 6502ef90a9
C2	AeST cosmology, native H_0	COMPUTED + DATA-GATED	9/9	ac4b30ff62 / 2dd388643a
C3	AeST scales vs $L_\star$	SCOPED NO-GO	9/11	c16ef33ae5 / fd7196584c
C4	Khronon parent	EMBEDS WITH EXTRA TERM	8/8	b0ea16ecf1 / 5f4f845554
C5	health	CONDITIONALLY HEALTHY	7/10	f002ce909a / f3e7d4d661
C6	Cassini, literal form	UNDECIDED (frozen)	7/8	1b2ced55a4 / ae428dc252
C7	native CMB code	RUNNABLE NOW	7/7	milestones to b0ced915c8
C8	instantaneous sector	WELL-POSED + UNCONSTRAINED	8/8	f2399aaf5e / aa113a5e96
D10	native $L_\star$ floor	CRISP FINDING	10/10	e3333cb274, A1 ecec1ce9e7 / 3ac0573063
E6	native catchment	CRISP FINDING	5/5	c42925d9dc, A1 5c0669a33d / 0bea588820

- **The free function.** $J = J_x = \Lambda - 2a_0^2[y - \ln(1+y)]$, $y = \sqrt{Y}/a_0$, gives $1 + J' = y/(1+y)$: AQUAL with $\mu = x/(1+x)$ exactly (C4, sympy). But $J_x/\sqrt{Y} \to -2a_0$ at large Y, so it lies outside Blanchet and Skordis's admissible class, which demands $J \to$ const. POSTULATED.
- **The addition.** **P** promoted to an independent spatial covector ($n \cdot P = 0$) with one stiffness term $-(\gamma_L/2)(D_i P^i)^2$, $\gamma_L = K L_\star^2/2$: one new field and one new constant. The pure Khronon cannot screen the Sun: with J_x the planets see the full a_0, 1.08×10^4–8.07×10^4 times their bounds; with a turn-off J the AQUAL Q_2 still fails by at least 2.44× (canonical) and 2.85× (alternative) (C4, b0ea16ecf1 / 5f4f845554).
- **Static limit.** $\phi = \psi$ (lensing sees the dynamical potential) and $\nabla\cdot[(1+J')\nabla\psi] = 4\pi G\rho$ for any mass distribution: the field equation is AQUAL, not QUMOND, matching PAPER58's E2. The curl that AQUAL needs lives in the unconstrained auxiliary P^* ($|\nabla \times P^*|/|\nabla P^*| = 0.126$ on a two-mass test), not in the foliation.
- **Relation to AeST (C1, C3).** AeST (Skordis & Złośnik 2021) reproduces AQUAL with $\mu = x/(1+x)$, the clock term and the curl half of the stiffness exactly, but the longitudinal stiffness $\gamma_L(D \cdot P)^2$ is identically zero in AeST and in any first-derivative extension; it needs second derivatives of AeST's scalar (C1, d9a9a17f4e / 6502ef90a9). No term in AeST's class screens small systems (C3, SCOPED NO-GO, c16ef33ae5 / fd7196584c): AeST's mass term acts at large r and in the opposite direction. So $L_\star$ is a new constant of the covariant theory.
- **Relation to DSF-21's A33.** A33 is this theory with $\eta > 0$. With the Khronon's curvature term ($\beta = 0$), $\eta > 0$ gives a gradient instability wherever the field is nonzero; in the Solar System at $k = 1/\mathrm{AU}$ the rate is 83.3 /yr (canonical) and 95.09 /yr (alternative). So η is dropped (C5).
- **A closure identity (C4).** If $J(\infty) = 0$ (Blanchet–Skordis 2025), the action gives $\Lambda c^4/a_0^2 = 2F(y_t)$, twice the record's closure, so the turn-off scale would be 33.94 (canonical), 23.70 (alternative) or 33.98 ($\kappa = \frac{1}{2}$). This applies only to the turn-off branch, which fails Cassini anyway, and it does not select κ.

3 Health (C5)

Criteria f002ce909a, result f3e7d4d661. Verdict CONDITIONALLY HEALTHY, and only in the literal form.

- **The literal form.** No ghost and no gradient instability on every background tested where the frozen-coefficient (WKB) expansion is valid; only the two tensor modes propagate, at c; **P** and the khronon scalar obey elliptic constraints; the $\omega = 0$ mode has positive energy at every k; the FRW dust has $c_s^2 = (\bar{Q} - 1)/\bar{Q}$. Conditions: $\gamma_L > 0$; W convex (the record's W meets this at every field strength); a positive dust charge; $\beta = 0$ and $\eta = 0$ exactly.
- **What fails.** $\eta > 0$ with $\beta = 0$: ill-posed (above; the stability boundary is the record's $\beta > 2\eta a_0^2/K$, confirmed at $\beta = 1.48707 \times 10^{-19}$). $\beta > 0$ with the $K(Q)$ dust: an infrared gradient band. AeST plus the stiffness term is second best: its $\omega = 0$ mode has negative energy at every k in vacuum, and the stiffness term gives an unbounded speed.
- **Checks.** 7/10, exit 1. Three frozen checks fail and are kept: H5 (a weak-field factor fails at neutron-star compactness), H6 (all failing points lie where the background is not local; this reading is post hoc) and H9 (the $\beta > 0$ dust band, physics). Strong coupling is far below every sub-Hubble scale.

4 Well-posedness and causality (C8)

Criteria f2399aaf5e, result aa113a5e96. WELL-POSED + UNCONSTRAINED, scoped to the weak-field leaf problem; 8/8 scored checks.

- **Well posed.** On each leaf the **P** and lapse equations are the Euler–Lagrange equations of one strictly convex functional, so the solution exists and is unique on a bounded domain for every source below the cap ($\gamma_L > 0$). Because the action has no $\dot{N}$ or $\dot{P}$, the time-dependent leaf problem is exactly the static one. OPEN: the infinite-volume theorem, the coupled system with GR's tensor sector and the dust, and strong fields.
- **Criterion A is violated; criterion B is met.** The response is a real, gauge-invariant infinite-speed influence on the khronon's leaves. It never goes backward in τ, so the single-global-time criterion holds. Its size: moving 1000 kg by 1 m changes the force 1 km away by $1.2 \times 10^{-46}\,\mathrm{m\,s^{-2}}$ at $L_\star = 24$ pc.
- **No existing data constrain it.** Binary pulsars: no radiative channel, and the conservative correction is at most 3.2×10^{-14} of the double pulsar's precision. Preferred frame: an α_2-equivalent of 1.7×10^{-7} of the bound (this also corrects the record's A22 D5, which missed the leaf tilt). GW170817: $\Delta v/c \leq 6.2 \times 10^{-32}$.

5 Parameters

- a_0 **(κ): FITTED.** $\kappa_\Lambda = 0.417 \pm 0.095$ (CFG493); 0.414 ± 0.094 on a ΛCDM-free background (PAPER58, E4).
- $L_\star$**: FITTED, a new constant, lower bound only.** With the theory's own field equation in the Galactic field (D10, result 3ac0573063), the floor is $L_{\star,95} = 21.68$ pc (canonical) / 22.85 pc (alternative) with the stellar-population M/L; over the M/L priors and the Milky-Way bracket it spans 2.93–22.67 / 4.40–23.80 pc and straddles 20 pc. The record's upper edge is about 90 pc (PAPER58, D8).
- **The dust abundance: FITTED.** It is an initial condition, the khronon's conserved shift charge (C4) or AeST's (C2); nothing in the action fixes it.
- $G_{\rm cosmo}/G_N$**: bounded.** BBN (the record's A78): $G_{\rm cosmo}/G_N > 0.9611$. A Planck-lite emulation gives $\sigma = 0.027$ (C2, wave 2, 5e93769089). In AeST it can only be lowered, and the same BBN line gives $K_B < 0.0778$ (C2).
- u **(the retained fraction):** a measured, convention-dependent function of scale with no mechanism (PAPER58, D6, E1, E4).

6 Tests

6.1 Galaxies

The static limit is the law itself (AQUAL, $\mu = x/(1+x)$; C4, PAPER58 E2). Galaxy rotation curves therefore test the law, as in the companion notes; this theory adds nothing at galactic scales.

6.2 Cassini and the planets (C6, D7d)

Criteria 1b2ced55a4, result ae428dc252.

- **The literal form passes Cassini.** All 24 cells ($L_\star$ = 24, 50, 90 pc; both footings; four Galactic fields) pass Hees et al. (2014) and Park et al. (2026). At 24 pc $Q_2 = 0.93$–$1.86 \times 10^{-34}\,\mathrm{s^{-2}}$; at 90 pc 1.76–3.52×10^{-36}. The margin below the Hees line is 4.8×10^7 to 5.1×10^9.
- **Mechanism.** Inside $L_\star$ the Sun's field is a constant hedgehog $GM/L_\star^2\,\hat{r}$ with no quadrupole. Removing the curl stiffness (D7d used it) lowers Q_2 by 9–19%; it does not open a quadrupole.
- **The frozen rule reads UNDECIDED.** One validation check (C2) fails at 3.8×10^{-11} against a 10^{-12} line: rounding in $(\kappa^2/2 + 1) - 1$ at $\kappa = 0.01$. The identity itself is exact in sympy. The script exits 1 (7/8).
- **Planets.** The anomaly is the hedgehog $GM/L_\star^2$: at 24 pc Mars is at 0.173 of its bound, Saturn 0.035 and Earth 0.028.
- **The unscreened law, for contrast,** fails Cassini by 2.44–5.19× under AQUAL (PAPER58, E2).

6.3 Outer globular clusters (D10)

Each cluster was solved as a Plummer sphere in the Galactic field with the literal form's own equations. The unscreened law ($L_\star = 0$) is excluded in all 18 cells ($\Delta\chi^2 = 4.9$–22.8), and no finite $L_\star$ beats Newton. So the globulars give a lower bound only. Pal 4 and Pal 14 dominate; the bound is limited by the stellar M/L of Pal 4.

6.4 Dwarf satellites (E3, E6)

- The satellites sit $+0.89 \pm 0.13$ dex above the screened prediction (PAPER58, E3).
- **The framework's dynamics cannot fix a satellite's catchment** (E6, c42925d9dc / 0bea588820): it is set by the patch's initial overdensity and infall epoch. Median $z_{\rm req} = 11.28$; the required catchment is an ordinary 1.21σ fluctuation in an externally calibrated initial field.
- **The satellites' own chemistry** (leaky box with their [Fe/H], no ΛCDM input) bounds the catchment from below at a median 38.33 times the present baryons, so the median need is 0.211 of the supply; 6 of 36 tested systems are short at central values and none at 2σ. This is CONDITIONAL on $\zeta = 1$ winds and no pre-enrichment, and it still needs a mechanism that packs the cold mass inside the half-light radius, which the record does not have.

6.5 Binary pulsars and gravitational waves (C8)

See Section 4: no constraint; tensor speed exactly c.

6.6 ALMA (D4)

Expected unscorable on both dates (gate S/N 0.04–0.58 against a required 5). The plans are sealed; addendum E (ffb14e40bc) adds the $n = 1$ and $n = 2$ kernels as labelled parallel readings beside the $\nu_{\rm mono}$ verdict kernel, and the harness has the kernel switch (92fa9f9471).

7 Sealed predictions for data still to come

- **Gaia DR4 wide binaries (20 kAU).** Screened: $\gamma_v - 1 = 8.1 \times 10^{-6}$ / 1.9×10^{-6} / 5.8×10^{-7} at $L_\star = 24$ / 50 / 90 pc (C6). Unscreened law-first (AQUAL): 1.167 (canonical) / 1.199 (alternative) at the primary external field. Scored on the whole-catalogue $\hat{\gamma}$ under the owner's pre-data $\sigma_{\rm sys}$ rule (pre-DR4 note amendment A3, 3d82af0a7e): definition (b) primary, (a), (e), (g2) must agree, else UNDECIDED; footings scored separately. Expected power at 30,000 pairs under (b), from DR3 (non-scoring): 3.21–3.96σ canonical.
- **ALMA U4** (public 2026-11-06) and **GS4** (2026-11-28): frozen CFG806/CFG809 plans plus addendum E; expected unscorable.

8 Pending

- **Native CMB: pending.** Lane C7 patched CLASS v3.3.4 with AeST's linear equations, derived from the action; the patched code reproduces pristine CLASS in the exact dust limit to 2.2×10^{-6} and passes its validation (7/7, milestone 4, b0ced915c8). No native spectrum has been run or confronted with data. When it is, it can test only the dust sector and $G_{\rm cosmo}/G_N$: a_0 is absent from the background and from linear order (C2: the invariant that carries a_0 vanishes on FRW at first order), so the CMB cannot test the galactic law.
- **Native cosmic shear: pending.** The record's canonical settling rules are excluded by raw KiDS (CFG596, $\Delta\chi^2 = +68.7$). The pre-registered native 1024^3 rerun (CFG960) was restarted under amendment A6 (f72bef68ea; a bit-identical lower-memory engine) and was running past step 90 of 150 when this note was written; it has no verdict on either footing.

9 What this is not

- Not a UV completion: an effective theory with a preferred foliation.

- Not derived constants: κ, $L_\star$ and the dust abundance are fitted; the free function J_x and the stiffness term are postulated.
- The CMB cannot test the galactic law at linear order.
- Criterion A (no influence faster than light) is violated, by a real signal too small for any test identified.
- The dwarf supply is conditional (chemistry assumptions, no packing mechanism).
- No particle species is added: the extra mass is the cold-energy sector, and the dust here is a field sector. No result here is evidence for the framework over its rivals.

Disclosures. C3 scores 9/11 with two frozen fails kept and a post-result amendment (6dba29d773). C5's three frozen fails (H5, H6, H9) are kept; its first complete run (6/10) had two code bugs, fixed with no pass line changed. C6's frozen verdict is UNDECIDED (float cancellation); its earlier executions and a 1.04 GB mutation run are disclosed. C8's first run (6/8) failed on rounding and a solver limit, and peaked at 2.3 GB; fixes changed no pass line. D10's amendment A1 is pre-result. E6's amendment A1 is pre-result and A2 post-result. The Khronon equations come from the arXiv HTML via the C4 lane and are not verified on disk; the AeST sources are on disk (C2 provenance). The pre-DR4 note's amendment A3 repeats a statement about the alternative footing that PAPER58 v1.1 withdrew; the scoring rule does not depend on it.

Repository: `covariant_push_2026_10_22/` (C1–C8, `THEORY_V3_DRAFT.md`); `doors_push_2026_10_22/` (D10, E6; D2, D4, D6, D7d, E1–E4 via PAPER58); `prep_2026/gaia_dr4_prep/` (pre-DR4 note, A1–A3). Companion notes: PAPER58 v1.1 (doi:10.5281/zenodo.23579379), PAPER57 (doi:10.5281/zenodo.23532885), PAPER49 v1.2 (doi:10.5281/zenodo.23533323). Theory: L. Blanchet, C. Skordis, JCAP 11 (2024) 040 (arXiv:2404.06584); L. Blanchet, C. Skordis, arXiv:2507.00912 (2025); C. Skordis, T. Złośnik, PRL 127, 161302 (2021) (arXiv:2007.00082); M. Milgrom, arXiv:1605.07458. Bounds: A. Hees, W. M. Folkner, R. A. Jacobson, R. S. Park, PRD 89, 102002 (2014); R. S. Park et al., PRD 114, 024066 (2026). AI assistance: Anthropic's Claude (lanes, review, drafting).